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Answer review

Q1 Single correct Correct 1 / 1

What does Energy Payback Time (EPBT) represent in a solar energy system?

A Time required for the system to generate energy equal to its embodied energy — Your answer — Correct answer

B Total operating life of the system

C Time required to install the system

D Time required to replace the system

Q2 Single correct Not attempted 0 / 1

What does Electricity Production Factor (EPF) mainly indicate?

A Ratio of lifetime energy generation to embodied energy — Correct answer

B Ratio of system cost to electricity tariff

C Ratio of battery voltage to current

D Ratio of solar radiation to temperature

Q3 Single correct Correct 1 / 1

What does Life Cycle Conversion Efficiency (LCCE) indicate?

A Ratio of lifetime useful energy output to total life cycle energy input
Your answer Correct answer

B Ratio of inverter voltage to PV voltage

C Ratio of installation cost to maintenance cost

D Ratio of module area to roof area

Q4 Single correct Correct 1 / 1

A solar system has an embodied energy of 5000 kWh and produces 2500 kWh of energy per year. What is its EPBT?

A 1 year

B 2 years
Your answer Correct answer

C 5 years

D 10 years

Q5 Single correct Not attempted 0 / 1

A PV system has an embodied energy of 6000 kWh and generates 3000 kWh of electricity annually. What is its EPBT?

A 1 year

B 2 years
Correct answer

C 3 years

D 6 years

Q6 Single correct Correct 1 / 1

A solar PV system generates 30000 kWh during its lifetime and has an embodied energy of 5000 kWh. What is its EPF?

A 5

B 6
Your answer Correct answer

C 10

D 25

Q7 Single correct Correct 1 / 1

What is the main energy benefit of a solar daylighting system in a building?

A It reduces the need for artificial lighting during daytime
Your answer Correct answer

B It increases electricity consumption during daytime

C It requires a battery for all applications

D It converts all sunlight into heat

Q8 Single correct Correct 1 / 1

For a solar daylighting system, a lower EPBT generally indicates:

A Faster recovery of the embodied energy
Your answer Correct answer

B Higher electricity consumption

C Longer installation time

D Lower availability of sunlight

Q9 Single correct Correct 1 / 1

A solar daylighting system has an embodied energy of 1200 kWh and provides an annual energy saving of 600 kWh. What is its EPBT?

A 1 year

B 2 years
Your answer Correct answer

C 3 years

D 6 years

Q10 Single correct Correct 1 / 1

For a solar PV plant, EPF generally increases when:

A Lifetime electricity generation increases for the same embodied energy

Your answer Correct answer

B Lifetime electricity generation decreases

C Embodied energy increases without any change in generation

D The PV system remains disconnected

Q11 Single correct Correct 1 / 1

A solar PV plant has an embodied energy of 8000 kWh and a lifetime electricity generation of 80000 kWh. What is its EPF?

A 5

B 8

C 10

Your answer Correct answer

D 20

Q12 Single correct Correct 1 / 1

Which combination represents a hybrid solar system using electrical and thermal solar technologies?

A Solar PV plus Solar Thermal

Your answer Correct answer

B Solar PV plus Diesel Generator

C Solar Thermal plus Wind Turbine

D Solar PV plus Grid Supply

Q13 Single correct Correct 1 / 1

Which combination represents a hybrid solar system that uses solar heat and natural daylight?

A Solar Thermal plus Solar Daylighting
Your answer Correct answer

B Solar PV plus Battery

C Solar PV plus Wind Energy

D Solar Thermal plus Diesel Generator

Q14 Single correct Correct 1 / 1

Which combination represents a hybrid solar system that provides solar electricity and natural daylight?

A Solar PV plus Solar Daylighting
Your answer Correct answer

B Solar Thermal plus Battery

C Solar PV plus Diesel Generator

D Solar Thermal plus Wind Energy

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